1. Describe one result you found interesting doing your task.

One result I found interesting while preparing our group's slides was how strongly heat transport changed with the temperature gradients. Increasing both the electron and ion temperature gradients from 1.5 to 3.5 raised the predicted deuterium ion heat flux from about 4.8 to 60.6, a 12.6-fold increase. The gradients increased much less than the heat flux, which showed a strong nonlinear response over the sampled cases. We also found that increasing the safety-factor gradient raised both ion and electron heat fluxes. These patterns helped me understand why both temperature gradients and magnetic geometry matter when building a surrogate for TGLF.

2. Describe one thing you struggled with doing your task.

The biggest difficulty was making sure the comparisons in our slides were meaningful. The dataset contains different ion species and several changing plasma parameters, so comparing arbitrary cases could mix together multiple effects. I had to pay attention to which inputs stayed fixed and which ones changed. For example, both temperature gradients change together in that scan, so the plots cannot tell us the separate effect of each gradient. It was also important to interpret the heat fluxes correctly: they are TGLF SAT2 predictions in gyro-Bohm units, even though the starter database is named SAT3.

3. Identify one choice that you made in your investigation. Why did you make this choice?

One choice I made was to focus my contribution on temperature gradients and the safety-factor gradient, complementing my groupmate's slides on density gradients and collisions. I chose matched scans so that each graph would have a clear interpretation. For the safety-factor-gradient comparison, the other inputs remain fixed, which makes the change in predicted heat flux easier to connect to that parameter. A broader overview of all cases would also have been useful, but these focused comparisons gave us clearer patterns to explain within a short group presentation and helped identify candidate inputs for our later modeling work.